Seclusion EDA Findings – CRDC 2020-21 (COVID year)

Prepared for: Adhya Putta · October Milestone · Scope: 2020-21 school-level data, used as the COVID check against 2017-18 and 2021-22

Bottom line

2020-21 shows the same structure as the other years: seclusion is rare (6.0% of reporting schools), skewed, and dominated by state. But **22% of schools have no usable outcome, and the gap is concentrated in a few states**, so this year is a sensitivity cohort, not a primary training year. Within a balanced panel of the same schools, seclusion **dipped in 2020-21 and rebounded in 2021-22**. The same target definition as the other years still applies.

Data completeness

- All 16 audit counts in the cleaning script reproduce exactly. 97,575 schools; 76,082 (78.0%) are analysis-ready.
- 21,493 schools lose the outcome: data-quality suppression (-11) 14,166 (14.5%), structural skips 6,575 (including 4,700 virtual-only skips), and 288 missing.
- **Suppression is state-driven:** Hawaii 100%, Florida 54%, Nevada 51%, North Carolina 42%, Maryland 38%, New York 36%. Those states averaged 2.7% any-seclusion in 2017-18 versus 9.5% for states with 90%+ valid outcomes, so the analysis sample over-represents high-reporting states.
- Virtual-only schools (5,189) skipped the module: only 342 have a valid outcome.
- The file has no readiness flag, so the analysis-ready subset must be derived.

What the data show

- **Distribution:** 94.0% of schools report zero. Among the 4,542 that report any, the median is 4 instances, the 90th percentile 31, the maximum 1,714. The top 100 schools hold 32.1% of instances.

- **State dominates:** Iowa 29.3%, Wisconsin 27.8%, Connecticut 25.9%, Maine 22.9% report any; California, Texas, Massachusetts and New York are near or under 1.1%. State alone gives ROC-AUC 0.80, and adding it lifts a baseline model's average precision from 0.16 to 0.32.
- **Concentration:** 89 districts hold half of all instances, and the top 10 hold 18.0%, the most concentrated of any year. One Nebraska school reports 1,714 instances for 109 students.
- **School type:** special-education schools 20.5% any (vs 5.7%); justice facilities 9.7%; alternative 1.7%, charter 1.5%, magnet 4.7%. Elementary 8.7%, middle 4.1%, high 2.3%.
- **Population:** %IDEA is the clearest positive correlate (2.6% any under 5% IDEA, 8.0% at 20-30%); %504 is the strongest negative (1.4% at 10% or more). All rank correlations are weak ($|\rho| \leq 0.08$).
- **Staffing:** non-monotonic and confounded. After adjustment, psychologists and social workers are positive; security guards, law-enforcement staff and nurses negative. 758 schools carry the cleaning outlier flag.
- **COVID instruction mode:** 87.8% of schools are coded hybrid, so the indicator has little resolution. In-person-only schools have lower odds of any seclusion than hybrid after adjusting for state and size (0.60), but the other groups are too small to trust.
- **Small schools distort rates:** schools under 30 students are 3.8% of schools but 43% of those with more than 100 instances per 100 students.

COVID check: 2017-18 vs 2020-21 vs 2021-22

- **Balanced panel** (67,775 schools valid in all three waves): any-seclusion went 7.71% → 6.13% → 8.27%. Elevated schools went 3.72% → 2.43% → 3.80%. Total instances fell 58% (121,009 → 50,917) and recovered to 89,552.
- **Selection-robust views agree:** naive prevalence 6.56% → 5.97% → 7.09%; post-stratified to the 2020-21 state mix 6.59% → 5.55% → 7.23%; the 20 high-validity states 9.30% → 8.09% → 10.13%. Each shows a dip of about one point.
- **Persistence stays high:** 38.1% of schools with any in 2017-18 also have any in 2020-21 (3.5% of the others); 58.3% of 2020-21 positives have any in 2021-22.
- **Where the dip sits:** hybrid schools (7.92% → 6.23% → 8.50%); in-person-only schools were flat (5.77% → 5.63% → 5.68%). This is association only.

Equity (sex and disability)

- Boys are 82% of secluded students.
- Students with IDEA are 13.9% of enrolled students but 75.2% of secluded students: 0.22 secluded students per 100 vs 0.012 for others (about 19x). IDEA boys are secluded at 0.288 per 100, about 52x the rate of non-IDEA girls.
- IDEA students were secluded 4.1 times on average vs 2.0 for others.
- Race could not be assessed.

Leakage

- Physical restraint: 26.9% of schools with restraint report seclusion vs 1.7% without (AUC 0.83 alone). Adding it to the baseline raises ROC-AUC from 0.73 to 0.87 and average precision from 0.16 to 0.43. Mechanical restraint is rare (0.8%) and weak but belongs to the same module. Exclude both.
- The seclusion components, the rate and the secluded-student counts are the outcome itself.

Recommended target definition

- **Primary, `any_seclusion`:** seclusion_instances ≥ 1. Prevalence 6.1% in the modeling frame.
- **Secondary, `elevated_seclusion`:** seclusion_instances ≥ 5 and rate ≥ 1 per 100 students. Prevalence 2.4% (1,756 schools).
- **Role of this year:** sensitivity or robustness cohort only. If pooled with other years, add a year indicator and a state effect (or state validity weights).
- **Modeling frame, if used:** analysis-ready, enrollment ≥ 30 (drops 2,864 schools), excluding the 758 FTE-outlier schools, with cross-validation grouped by district. Report virtual-only schools separately.
- **Not recommended:** raw rate or count as a regression target.

Candidate feature shortlist

- **Include:** log enrollment, grade band, special-education / alternative / charter / justice flags, %IDEA, %504, %EL, psychologists and social workers per 100 students, log district enrollment.

- **Optional:** magnet flag, counselors, nurses and teachers per 100, security staff per 100 (reverse-causality caution), COVID instruction type (2020-21 only).
- **Second model only:** state (or region).
- **Drop:** raw teacher and counselor FTE, `LEA_SCHOOLS` (0.96 with district enrollment).
- **Exclude as leakage:** the seclusion components and rate, secluded-student counts, and all physical and mechanical restraint counts.

Data issues for the cleaning owners

- **High:** 22% of schools have no usable outcome, and -11 suppression is concentrated by state; provide state-level validity weights.
- **High:** virtual-only schools skipped the module (-13); virtual vs in-person cannot be compared fairly.
- **High:** the file is named analysis-ready but has no readiness flag and is not filtered.
- **Medium:** 758 implausible-FTE schools retained (for example 833 nurses at one school); 82 schools with more instances than students; COVID indicators have limited resolution; schema, small-enrollment flag and grade band differ from the other years; race and other discipline columns not carried through.
- **Low:** `COMBOKEY_stored` left in the output; `seclusion_rate_per_100` naming; law-enforcement FTE 5.7% missing.

The full list with row counts is in `data_issues_202021.csv`.

Open items

1. Provide state-level validity weights, or a documented rule for handling -11 suppression, before any pooled model.
2. Carry race and other discipline columns through cleaning, then rerun the equity and leakage sections.
3. Team decision: whether 2020-21 is used only as a sensitivity year, the minimum enrollment of 30, and whether the headline model includes state.